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What if Hurricane Katrina Hit in 2020? The Need for Strategic Management of Disasters

Part IV: The Future of Strategic Management

What would another Hurricane Katrina bring in 2020? Is government at all levels ready for such an event? This essay argues that learning to manage actions supporting disaster response and recovery operations consistently and effectively requires strategic, not reactive, thinking. It adopts a framework based in issues focused thinking, externally focused management, and opportunity-seeking behavior to examine the complex mix of individuals, organizations, and actions involved in response and recovery. The authors conclude that strides have been made since Katrina, but the development of greater strategic capacity is a necessary prerequisite to effective disaster management in the future.

Wise and fortunate indeed is that community that has so analyzed its problems and needs, and has so prepared to make use of catastrophe should it come by plans for carrying out programs of improvements, that the aftermath of tragedy finds its victims as well cared for as humanly possible and in addition some tangible new advance in the equipment and life of the community.

—John Gaus, 1947

Scores of reports, newspaper accounts, documentaries, and academic articles have been produced about the catastrophic U.S. hurricane of 2005, Hurricane Katrina. Many of these are empirical, but some are polemical, bemoaning the work of government officials in the disaster—and with good reason. There is little disagreement that government agencies at all levels failed in some aspect to protect one of the affected cities, New Orleans, before the event and botched the response to, and recovery from, the event after the hurricane struck. Certainly, the shock of Katrina reminded everyone in government of the valuable lesson that management, organization, and, especially, leadership matter, particularly in times of crisis. Catastrophe has a

way of exposing weaknesses in an administrative structure, and tragedy leads all to reevaluate their circumstances, be they organizations, communities, or individuals.

What would another Hurricane Katrina bring in 2020? Is government at all levels ready for such an event? Or is the worst yet to come? (Kettl 2006). In the following sections, we argue that successful emergency preparedness, response, and recovery in the future are largely functions of the degree to which government leaders possess strategic management capacity. Such management is an ongoing

process integrating planning that is *issue focused*, *externally oriented*, and *opportunity seeking* (Koteen 1997). We do not sound the alarm that is so common among post-9/11 and Katrina commentators, but rather offer a strategic framework to enhance our understanding of the importance of management and leadership before and during a Katrina-like event occurring in the year 2020. We cannot point to the characteristics of what an “ideal” response and recovery for any given situation or locale might look like in this short essay—if indeed such characteristics exist and could be identified—but we do offer perspectives on the critical managerial components of an emergency management strategy. Our observations are grounded in the notion that a “disaster occurs when the demands of the system are greater than the capabilities of the community to meet them” (Simpson 2008). Put another way, the impact of a disaster “agent” (such as a hurricane) alone does not determine whether an actual disaster occurred; it is also a question of whether the community has the requisite strategic capacity to effectively mitigate, address, and cope with the incident.

Because much of the blame for the overall ineffective response to Katrina is placed on the Federal Emergency Management Agency (FEMA), as the nation’s

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coordinating arm for natural disasters and catastrophes, we use FEMA as a lens through which strategic capacity can be assessed and developed. Nonetheless, because basic federal law and practice have never abandoned the principle that initial responsibility rests with first responders and local management, we also demonstrate the importance of the relationship between all levels of government, nongovernmental organizations, and the general public, and how planning and support at the federal level should fan out through the intergovernmental system. We can never be 100 percent sure whether any city, region, state, or nation is prepared for any given unique catastrophic event, but certain managerial principles can be established that will enable a government scarred by Katrina to do its best and to more effectively mitigate and plan, and carry out its response and recovery obligations in the future. We conclude by arguing that, five years after Katrina, progress is being made, but the gains need to be consistently supported and maintained, with substantially more to be accomplished. But first, we examine the state of emergency management in the United States in the years immediately preceding Hurricane Katrina.

The State of Emergency Management Pre-Katrina

It is important to note at the outset, particularly within the context of the failures of Hurricane Katrina, that FEMA, like many other large agencies such as NASA or the U.S. Postal Service, has experienced cycles of failure, reform, success, deterioration, failure, and so on, for decades. Thus, reform initiatives are no stranger to FEMA or to the federal government in general. In August 2005, the focus of reform efforts within the U.S. government was to support the “war on terror.” Federal efforts, strategic and operational plans, reorganizations, and budgets established after the terrorist attacks on September 11, 2001, reflected this focus. During this time, frequently occurring natural and technological hazards quickly became the “poor cousin” to the potential for catastrophe resulting from international terrorism. Prior to 9/11, under the George W. Bush administration, FEMA, which was primarily associated with the federal coordination of large-scale responses to natural and technological hazards, lost its cabinet position, was given an administrator with virtually no emergency management experience, and faced a one-third cut in its proposed budget (Bea 2002; Hogue and Bea 2006).

The emergency management field was a patchwork of relationships and requirements, and as broad as the type of risks that society faced. In addition to dealing with natural and technological disasters, public health threats, environmental issues resulting in economic loss, cyberterrorism, and even astronomical issues (e.g., the possibility of asteroid and meteor strikes) were all on the plate of the field. It was a challenge, even in an all-hazards environment, to find common ground for discussions about particular types of disasters, as well as to accommodate the wide variety of professional interests from civil engineering to health and human services to psychological counseling. Guidance and “best

practices” emanated from both governmental and nongovernmental organizations, but rarely did one consistent set exist for a given area (Waugh 2007).

Within two years after 9/11, FEMA had been absorbed into the newly cobbled together Department of Homeland Security along with more than 20 other organizational entities, many of the lessons identified as a result of large-scale emergency exercises had been generally forgotten or ignored, and a substantial number of physical infrastructure projects nationwide had been placed on hold. Nonetheless, actual hazard response seemed to improve; not since Hurricane Andrew in 1992 had a large-scale natural disaster been complicated by poor response.

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At the time of Hurricane Katrina, on August 29, 2005, emergency management and program content and fiscal support varied wildly by region and locality; state and local programs had barely had a chance to reorganize in support of the new war on terror expectations; and some of the people who filled emergency management–related positions had rather eclectic educational and professional backgrounds. Many brought with them ideas, standards, and communication and organizational skills influenced by their primary discipline or field. The content of local emergency plans reflected this diversity, ranging from those composed of simply an “evacuation plan” (a map and a description of major area routes) to a more mature, fully developed plan, complete with an identification

and analysis of likely risks, personnel qualifications, memoranda of understanding of support between organizations, an evacuation plan based on likely hazards and computer modeling, and requirements for training, drills, and periodic exercises (Edwards and Goodrich 2007; Grant 1996).

The unevenness in the maturity and capacity of local emergency administrative infrastructures not only resulted in legitimate regional differences, but also translated into an unevenness in strategic capacity, with some organizations being highly competent, while others were unable to perform even the most routine tasks (Sutphen and Waugh 1998). This, along with diversity in organizational entities of different sizes and types working together, tended to result in fragmentation attributable to a divergence of federal, state, and local government approaches to emergency management and differing views of emergency management based on individual frames of reference, leading to conflicts across peer organizations and among neighboring geographic jurisdictions (Drabek 2003; McEntire and Dawson 2007).

Furthermore, FEMA’s Federal Response Plan, an operational plan that laid out the individual roles and responsibilities of federal responders, had been replaced in the post-9/11 environment with the National Response Plan (NRP) and the National Incident Management System. The NRP, in particular, attempted to impose federal expectations, not only on members of the federal family, but also on state and local entities in their efforts to deal with an

all-hazards world; to address the lack of standardization of local and state emergency management organizations; and, more importantly, to consistently address emergency management issues associated with a terrorist attack.

At the time of Hurricane Katrina, the NRP was in place yet remained largely untested. It had been used for smaller emergencies prior to Katrina, but the hurricane marked the first time the NRP was used for a catastrophic event. Complicating matters further, FEMA headquarters and its 10 regions differed in their philosophies and cultures. Headquarters was responsible for setting policy, but each region was responsible for interpreting and implementing that policy. With political appointees heading up each FEMA region as well as headquarters, and FEMA overall weakened by its absorption into the Department of Homeland Security, regional administrators felt free to act independently, and to rely on their own judgment and that of the states and locals within their region. Conflict among regions and even between managers within a FEMA region and at headquarters occurred frequently (Olasky 2006; Sylves 2006).

Strategic Emergency Management for 2020

It is hardly a surprise, then, that preparation for and the response to Hurricane Katrina resulted in failure. Shortly after the storm struck, Sandra Schneider asserts, “some of the major problems encountered during the response to Katrina stemmed from breakdowns in the administrative elements of the governmental emergency management system. Effective bureaucratic agencies are characterized by well-established procedures, effective leadership, and clear objectives. An early assessment suggests that problems associated with all three of these bureaucratic characteristics weakened governmental efforts to respond quickly and effectively” (2005, 515). State and local governments in Louisiana and Mississippi were simply overwhelmed following Hurricane Katrina. Failures and subsequent recommendations by the federal government are documented in reports issued by the U.S. House of Representatives, the U.S. Senate, and the White House, the last of which focused on the federal administrative response and offered a total of 125 recommendations in a multitude of areas, including review and revision of the NRP; increased involvement of the private sector and other nongovernmental organizations; the education and empowerment of citizens; and further integration and coordination of federal, state, and local efforts (U.S. House 2006; U.S. Senate 2006; White House 2006). However, little attention was given to strategic thinking and acting, and how incorporating strategic issues management into the bureaucratic process as well as intergovernmental coordination and collaboration might improve the management of future catastrophes.

Issue-Focused Management

The use of strategic issues management, which addresses the primary strategic concerns of an organization or organizational system, is important in emergency management because of the occurrence of unexpected and surprise events. Koteen (1997) argues that the criteria for identifying strategic issues include assessing (1) the current or potential impact on organizational performance, (2) the perceived urgency in significant consequences, and (3) the impact or interdependency with other issues. For more

than three decades, emergency management has been described in terms of a comprehensive, systems-based approach to activities comprising four management phases: mitigation, planning and preparedness, response, and recovery. However, most of the academic and popular literature has focused on one phase—emergency response—even as Hurricane Katrina exposed weaknesses in mitigation and recovery. Effective strategic management of disasters necessarily must focus on all issues and phases.

Historically, major hurricanes have left in their wake fresh congressional enactments that enlarged the federal role in disaster response, and Katrina was no different. One of the first major casualties of the emergency management infrastructure overhaul was the National Response Plan. It was deemed difficult to understand and implement; at more than 400 pages, overly complicated and bureaucratic; and generally confusing (Birkland and Waterman 2008). Its development had effectively excluded the participation of nonfederal stakeholders, such as states and localities, and suffered from limited comment periods for all parties, including federal partners and the public.

In response to the post-Katrina legislation and perceived problems with the implementation of the NRP, the Department of Homeland Security substantially revised the plan, with more involvement from stakeholders, and reissued it in 2008 as the National Response Framework. In its final form, the National Response Framework is part of an overall national strategy for homeland security, attempting to build adaptability and flexibility into response efforts by setting up a general framework rather than a specific prescription. Its intent is to provide the doctrine and guiding principles for a unified response at all levels of government for all types of hazards, but to give users some latitude as to how they choose to respond.

The framework reflects the reality that all emergencies are essentially local, and it works hand in hand with the National Incident Management System, which was also updated and revised in 2008. The National Incident Management System provides the template for the *management* of incidents, while the National Response Framework provides the structure and mechanisms for national-level *policy* for incident management. Working together, they are intended to provide a systematic, proactive approach to guide departments and agencies in all three sectors to work seamlessly to respond to the effects of incidents. The issue of ineffective disaster response, as reflected in one of its *fixes*, has thus been exposed as a failure of public management.

FEMA is also attempting to effectively address the strategic issue of recovery from disasters. The National Disaster Recovery Framework (in draft form at the time of this writing) is intended to move beyond the scope of the National Response Framework. The draft National Disaster Recovery Framework offers a spectrum of recovery actions that begins, in the ideal situation, before a disaster strikes. It states that postdisaster recovery activities should begin, or should be acknowledged as beginning, in the early stages of response, and that actions to support recovery activities should

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be built into governmental operations at all levels. Like the National Response Framework's philosophy of response, the draft National Disaster Recovery Framework recognizes the role of the federal government as a partner, coordinator, and facilitator, and notes that tribal, state, and local governments have primary responsibility for the recovery of their communities and play the lead role in planning and authority for managing all aspects of community recovery.

While the National Response Plan was justifiably deemed a failure and new general frameworks for response and recovery have been developed, many nongovernmental, street-level actors such as nonprofit agencies and social service organizations desire a strategic playbook of sorts based on past experience and designed to clarify expected actions during response and recovery. This desire for a step-by-step guide reflects (1) a failure of attention by local government planners to properly and thoroughly elucidate those involved, why, and how; (2) a gap in the extent to which learning occurs across and within communities that experience disasters; and (3) a need for street-level actors to know who in real time fulfills specific leadership and complementary roles. The frameworks guiding FEMA assume that federal, state, and local entities communicate and coordinate, and that localities establish and support over time the relationships, connectedness, and coordination essential for successful response and recovery. These assumptions belie the actual state of strategic capacity.

Externally Oriented Management

Disasters, by their very nature, typically exceed a single jurisdiction's (or entity's) ability or resources, and they almost never neatly contain themselves within a single city, county, or even state's boundaries. Drabek and McEntire note that "disasters, by their very disruptive and dynamic nature, create such significant demands on the affected community that well-executed, multiorganizational responses become not only necessary, but essential ... In addition, disasters often require the assistance of outsiders and multiple levels of government, thereby leading to multijurisdictional response operations" (2002, 206). Thus, the emergency management networks that characterize the field necessarily involve literally dozens of potential partners and sometimes thousands of volunteers. Petak explained long ago that, for emergency management, "understanding ... the intergovernmental complexities may be of greater importance to the success of activities than scientific knowledge, technical remedies, or their associated economic cost and benefits" (1985, 4–5).

The National Response Framework provides a clear endorsement of the idea of locally led coordination in emergency management (DHS 2008). It defines the duties of local emergency managers as having the day-to-day authority and responsibility in their jurisdictions for overseeing emergency management programs and activities. The document states that the local emergency manager "coordinates all components of the local emergency management program," and that includes coordinating the planning process; working collaboratively with other local agencies, which may include both nonprofit and for-profit organizations; and involving other organizations in planning, training, and exercises. Indeed, creating engaged partnerships is the first principle listed in the document. Even as local governments have often struggled to identify key local actors (Lindell et al. 1996), the National Response Framework emphasizes the idea that "com-

munities, tribes, States, the Federal Government, [nongovernmental organizations], and the private sector should each understand their respective roles and responsibilities, and complement each other in achieving shared goals" (DHS 2008, 4).

Our *understanding* of how to lead such an all-encompassing, multiorganizational process in emergency management is emerging, but *implementing* such principles on the ground is still a work in progress (Kettl 2007). Although it may be extremely challenging to address hardware and software problems that create obstacles to interoperability in communications and to design a consistent and unifying solution, it is far more difficult to effectively address the timing, frequency, content, and scope of interorganizational and interpersonal collaboration. One recent study of U.S. counties demonstrated that only when emergency managers found themselves in situations in which the problems facing their organization were increasingly severe did they reach out to collaborate with other entities and agencies (McGuire and Silvia 2010). It was found that the extent and level of external management is a function not only of the severity of the problem faced by the county, but also the level of capacity possessed by the manager and his or her agency. Furthermore, leadership actions that mobilize external support for the emergency management network and facilitate interaction within the network were found to positively relate to network effectiveness (McGuire and Silvia 2009). Central to the perceived effectiveness of a network is its established legitimacy in the eyes of its stakeholders and a demonstrated commitment by local officials to make disaster preparedness its highest priority (McEntire and Myers 2004).

However, the need to strategically manage actors within a networked framework suggests a fundamental tension between top-down, command and control management and more externally oriented bottom-up or lateral strategies (Harrauld 2006). As current FEMA administrator Craig Fugate asserts, "[Emergency managers] who get into this business and think that [their] job is to take charge, lead the response, and tell everybody how to do their jobs get very frustrated and don't last very long" (Pastula 2010). Thus, a necessary but sometimes difficult balance, the merging of hierarchy and collaborative networks, is present in emergency management.

As Moynihan (2005) shows, responses to man-made disasters can take place, paradoxically, through collaboration governed by command and control procedures. His case study of the outbreak of Exotic Newcastle Disease in the state of California describes the administrative capacity of a task force assigned with limiting and eliminating the disease. The task force operated much like a lateral, side-by-side collaborative arrangement, but it did so within the context of a top-down incident command system. The emergency response network was coordinated hierarchically, suggesting the existence of a "hierarchical network" (Moynihan 2008) or working "from the top down so that the system works from the bottom up" (Kettl 2006, 283). Such an external orientation is one of the pillars of strategic emergency management.

Opportunity-Seeking Management

Hurricane Katrina exposed severe weaknesses and disconnects within the emergency management infrastructure. Just as the strategic manager can attempt to remedy such weaknesses by looking

toward the external environment and identifying specific strategic issues to be addressed, so, too, does strategic management entail the pursuit of opportunities for improved performance. Although born of death and devastation and yet to be tested, FEMA perceived the failures of Katrina as an opportunity to design new frameworks that empower (if not directly supply capacity for) nonfederal governments to assume greater responsibility during and after disaster events. Capitalizing on such opportunities embodies the philosophy of strategic management.

There are countless opportunities to improve the different aspects of emergency management. For example, managers are finding that externally oriented management through networks and collaboration is not just a “soft” skill; it can also be facilitated by the innovative use of information technology. As Banipal demonstrates, integrated information management systems can impact and improve upon virtually all aspects of disaster response and recovery, “such as loss data accuracy, operational complexity reduction, purchasing and distribution of supplies, enhanced inter-agency communication and cooperation” (2006, 491). A relational database used to store and retrieve information can be used for various applications. For example, during an evacuation process,

[R]esidents should be urged to register at the site ... with specific information including name, address, number of family members evacuating and the city where family is planning to evacuate to. Based on the collected information, [a] city can determine how many people evacuated and how many still remain in the city. The information system can be used to generate statistical reports to show the pattern of evacuation and city to which [the] majority of people will evacuate. (Banipal 2006, 491)

A system similar to this could also be used for supply chain management, such as identifying the emergency resources needed at specific locations and the critical facilities within the supply chain (Hale and Moberg 2005).

FEMA administrator Fugate noted that “[i]t used to be that you needed a sophisticated GIS lab to do good mapping, incorporate satellite imagery and all that other stuff. Now you have so many products out there for the average person that you’re putting tools in the hands of responders that in many cases they never had access to. They can bring in data layers and real-time data from various sources and look at it in a GIS environment without needing a GIS shop to produce that” (Pastula 2010).

Several researchers are designing and attempting to implement information systems that can support large-scale collaboration. For example, the use of Web 2.0 technologies, including social networking sites such as Facebook and MySpace, and rapid communication systems, such as blogs and Twitter, offer an opportunity for “expert identification, resource aggregation and availability, and mass collaboration” on a scale not yet imagined or achieved (White et al. 2009, 7). One project, known as “Mega-Collaboration,” is based on a tool that “will surpass existing groupware and social

networking applications, providing easy entry, categorization, and visualization of masses of critical data,” along with facilitating the creation of ad hoc teams and coordinating large scale activities (Newlon et al. 2009, 1).

Conclusion

Emergency management is a complex mix of individuals, organizations, and actions, some of which will not be necessary in many types of disasters and emergencies and others that will be singularly and perhaps unexpectedly important, with rules and expectations that are most times ambiguous, and which tends to be highly interdisciplinary in nature. As a result, emergency managers and those who study the emerging field within public administration need to understand and appreciate a variety of disciplines. If viewed from one discipline or viewpoint, it is highly likely that important information, data, and insights will be missed, inhibiting our ability to completely understand individual catastrophes such as Katrina. For instance, it is important to understand not only the development and operation of bureaucratic organizations, but also interactions between organizations and networks within the organizations, communications, collective behavior, and individual social and psychological behavior in extreme environments (from the perspective of responders, and citizens and survivors alike).

Attempts to build strategic capacity are under way in the emergency management system, but such capacity is far from tested. Although imperfect, the responses to Hurricanes Gustav and Ike in 2008 provide some evidence that the United States is on the right track to respond to, and recover from, a disaster or catastrophe in 2020.

Attempts to build strategic capacity are under way in the emergency management system, but such capacity is far from tested. Although imperfect, the responses to Hurricanes Gustav and Ike in 2008 provide some evidence that the United States is on the right track to respond to, and recover from, a disaster or catastrophe in 2020. These two hurricanes devastated communities along the Gulf Coast and forced the evacuation of more than 2 million people. Coordination among federal, state, and local governments was found to be far improved (Fowler 2008a), and state officials reported that the federal response was good (Fowler 2008b). Thus, some lessons have been learned, but they need to be supported and communicated with others.

Learning to manage actions supporting disaster response and recovery operations consistently and effectively requires strategic, not reactive, thinking. From some perspectives, developing the requisite capacity to manage disaster operations can appear to be equivalent to the fate of Sisyphus. There needs to be developed a common understanding of what adequate planning and preparedness looks like in a variety of situations, along with an always imperfect yet effective response and recovery. But how is this to be done, short of offering only speculative guidelines, standards, and measures? (Henstra 2010). What does competence look like? What are we expecting to see? What are the assumptions of government and expectations of the public? The answers to these questions can be derived only through strategic activity.

We are encouraged by the strides made since August 2005 in the United States. However, success in emergency management is highly dependent on the continuation of time, funding, and the attention

of the general public and media, as well as consistency in political leadership. It is possible to have competently designed emergency response facilities, plans, and procedures, engaged and trained professional personnel, appropriately mitigated hazards, and an adequate amount of resources and material to support a catastrophic response and recovery, yet still fail. There is always the possibility that human error and hubris over recent successes can envelop an agency and make it complacent. Furthermore, mitigation plans, emergency operations and response plans, evacuation plans, recovery plans, and continuity plans cannot possibly account for all possible contingencies. Thus, the effective emergency manager in 2020 will attempt to forge strategic capability and develop the capacity of his or her organization to respond to forces that are beyond the organization's and the community's control.

References

- Banipal, Kulwinder. 2006. Strategic Approach to Disaster Management: Lessons Learned from Hurricane Katrina. *Disaster Prevention and Management* 15(3): 484–94.
- Bea, Keith. 2002. *Transfer of FEMA to the Department of Homeland Security: Issues for Congressional Oversight*. Washington, DC: Congressional Research Service. RL-31670.
- Birkland, Thomas, and Sarah Waterman. 2008. Is Federalism the Reason for Policy Failure in Hurricane Katrina? *Publius* 38(4): 692–714.
- Drabek, Thomas E. 2003. *Strategies for Coordinating Disaster Responses*. Monograph no. 61. Denver: Institute of Behavioral Sciences, University of Colorado.
- Drabek, Thomas E., and David A. McEntire. 2002. Emergency Phenomena and Multiorganizational Collaboration in Disasters: Lessons from the Research Literature. *International Journal of Mass Emergencies and Disasters* 20(2): 197–224.
- Edwards, Frances L., and Daniel C. Goodrich. 2007. Organizing for Emergency Management. In *Emergency Management: Principles and Practice for Local Government*, 2nd ed., edited by William L. Waugh, Jr., and Kathleen Tierney, 39–55. Washington, DC: ICMA Press.
- Fowler, David. 2008a. Who's in Charge Still a Disaster-Response Question Mark. *CQ Homeland Security*, September 22.
- . 2008b. Federal Response to Recent Hurricanes Termed “So Far, So Good.” *CQ Homeland Security*, September 23.
- Fugate, Craig. 2009. Post-Katrina: What It Takes to Cut the Bureaucracy and Assure a More Rapid Response after a Catastrophic Disaster. Testimony before the U.S. House Committee on Transportation and Infrastructure, Subcommittee on Economic Development, Public Buildings and Emergency Management, July 27. http://www.fema.gov/pdf/about/testimony/072709_fugate.pdf [accessed August 26, 2010].
- Gaus, John Merriman. 1947. *Reflections on Public Administration*. University: University of Alabama Press.
- Grant, Nancy K. 1996. Emergency Management Training and Education for Public Administrators. In *Disaster Management in the U.S. and Canada: The Politics, Policymaking, Administration and Analysis of Emergency Management*, 2nd ed., edited by Richard T. Sylves and William L. Waugh, Jr., 313–26. Springfield, IL: Charles C. Thomas.
- Hale, Trevor, and Christopher R. Moberg. 2005. Improving Supply Chain Disaster Preparedness: A Decision Process for Secure Site Location. *International Journal of Physical Distribution and Logistics Management* 35(3): 195–207.
- Harrauld, John R. 2006. Agility and Discipline: Critical Success Factors for Disaster Response. *Annals of the American Academy of Political and Social Science* 604: 256–72.
- Henstra, Daniel. 2010. Evaluating Local Government Emergency Management Programs: What Framework Should Public Managers Adopt? *Public Administration Review* 70(2): 236–46.
- Hogue, Henry B., and Keith Bea. 2006. *Federal Emergency Management and Homeland Security Organization: Historical Developments and Legislative Options*. Washington, DC: Congressional Research Service. RL-33369.
- Kettl, Donald F. 2006. Is The Worst Yet to Come? *Annals of the American Academy of Political and Social Science* 604: 273–87.
- . 2007. *System under Stress: Homeland Security and American Politics*. 2nd ed. Washington, DC: CQ Press.
- Koteen, Jack. 1997. *Strategic Management in Public and Nonprofit Organizations: Managing Public Concerns in an Era of Limits*. 2nd ed. New York: Praeger.
- Lindell, Michael K., David J. Whitney, Christina J. Futch, and Catherine S. Clause. 1996. The Local Emergency Planning Committee: A Better Way to Coordinate Disaster Planning. In *Disaster Management in the U.S. and Canada: The Politics, Policymaking, Administration, and Analysis of Emergency Management*, 2nd ed., edited by Richard T. Sylves and William L. Waugh, Jr., 234–49. Springfield, IL: Charles C. Thomas.
- McEntire, David A., and Gregg Dawson. 2007. The Intergovernmental Context. In *Emergency Management: Principles and Practice for Local Government*, 2nd ed., edited by William L. Waugh, Jr., and Kathleen Tierney, 57–70. Washington, DC: ICMA Press.
- McEntire, David A., and Amy Myers. 2004. Preparing Communities for Disasters: Issues and Processes for Government Readiness. *Disaster Prevention and Management* 11(5): 369–79.
- McGuire, Michael, and Chris Silvia. 2009. Does Leadership in Networks Matter? Examining the Effect of Leadership Behavior on Managers' Perception of Network Effectiveness. *Public Performance and Management Review* 33(1): 34–62.
- . 2010. The Effect of Problem Severity, Managerial and Organizational Capacity, and Agency Structure on Intergovernmental Collaboration: Evidence from Local Emergency Management. *Public Administration Review* 70(2): 279–88.
- Moynihan, Donald P. 2005. *Leveraging Collaborative Networks in Infrequent Emergency Situations*. Washington, DC: IBM Center for the Business of Government.
- . 2008. Combining Structural Forms in the Search for Policy Tools: Incident Command Systems in U.S. Crisis Management. *Governance* 21(2): 205–29.
- Newlon, Christine M., Mark Pfaff, Himalaya Patel, Gert-Jan de Vreede, and Karl MacDorman. 2009. Mega-Collaboration: The Inspiration and Development of an Interface for Large-Scale Disaster Response. Paper presented at the 6th International Conference on Information Systems for Crisis Response and Management, May 10–13, Gothenberg, Sweden.
- Olasky, Marvin. 2006. *The Politics of Disaster: Katrina, Big Government, and a New Strategy for Future Crises*. Nashville, TN: W Publishing Group.
- Pastula, Marty. 2010. FEMA Administrator Fugate Opens Up about Haiti, Ethics and Technology. *Emergency Management*, March 22. <http://www.emergencymgmt.com/disaster/FEMA-Administrator-Craig-Fugate.html> [accessed August 26, 2010].
- Petak, William J. 1985. Emergency Management: A Challenge for Public Administration. Special issue, *Public Administration Review* 45: 3–7.
- Schneider, Sandra K. 2005. Administrative Breakdowns in the Governmental Response to Hurricane Katrina. *Public Administration Review* 65(5): 515–16.
- Simpson, David M. 2008. Disaster Preparedness Measures: A Test Case Development and Application. *Disaster Prevention and Management* 17(5): 645–61.
- Sutphen, Sandra, and William L. Waugh, Jr. 1998. Organizational Reform and Technological Innovation in Emergency Management. *International Journal of Mass Emergencies and Disasters* 16(1): 7–12.
- Sylves, Richard T. 2006. President Bush and Hurricane Katrina: A Presidential Leadership Study. *Annals of the American Academy of Political and Social Science* 604: 26–44.
- U.S. Department of Homeland Security (DHS). 2008. National Response Framework. http://www.dhs.gov/files/programs/editorial_0566.shtm [accessed August 26, 2010].

- U.S. House Select Bipartisan Committee to Investigate the Preparation for and Response to Hurricane Katrina. 2006. *A Failure of Initiative*. Washington, DC: U.S. Government Printing Office. http://katrina.house.gov/full_katrina_report.htm [accessed August 26, 2010].
- U.S. Senate Committee on Homeland Security and Governmental Affairs. 2006. *Hurricane Katrina: A Nation Still Unprepared*. Washington, DC: U.S. Government Printing Office. <http://www.gpoaccess.gov/serialset/creports/katrinanation.html> [accessed August 26, 2010].
- Waugh, William L., Jr. 2007. Local Emergency Management in the Post 9/11 World. In *Emergency Management: Principles and Practice for Local Government*, 2nd ed., edited by William L. Waugh, Jr., and Kathleen Tierney, 4–22. Washington, DC: ICMA Press.
- White, Connie, Linda Plotnick, Jane Kushma, Starr Roxanne Hiltz, and Murray Turoff. 2009. An Online Social Network for Emergency Management. Paper presented at the 6th International Conference on Information Systems for Crisis Response and Management, May 10–13, Gothenberg, Sweden.
- White House. 2006. *The Federal Response to Hurricane Katrina: Lessons Learned*. Washington, DC: Office of the Assistant to the President for Homeland Security and Counter-terrorism. <http://georgewbush-whitehouse.archives.gov/reports/katrina-lessons-learned/> [accessed August 26, 2010].

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